

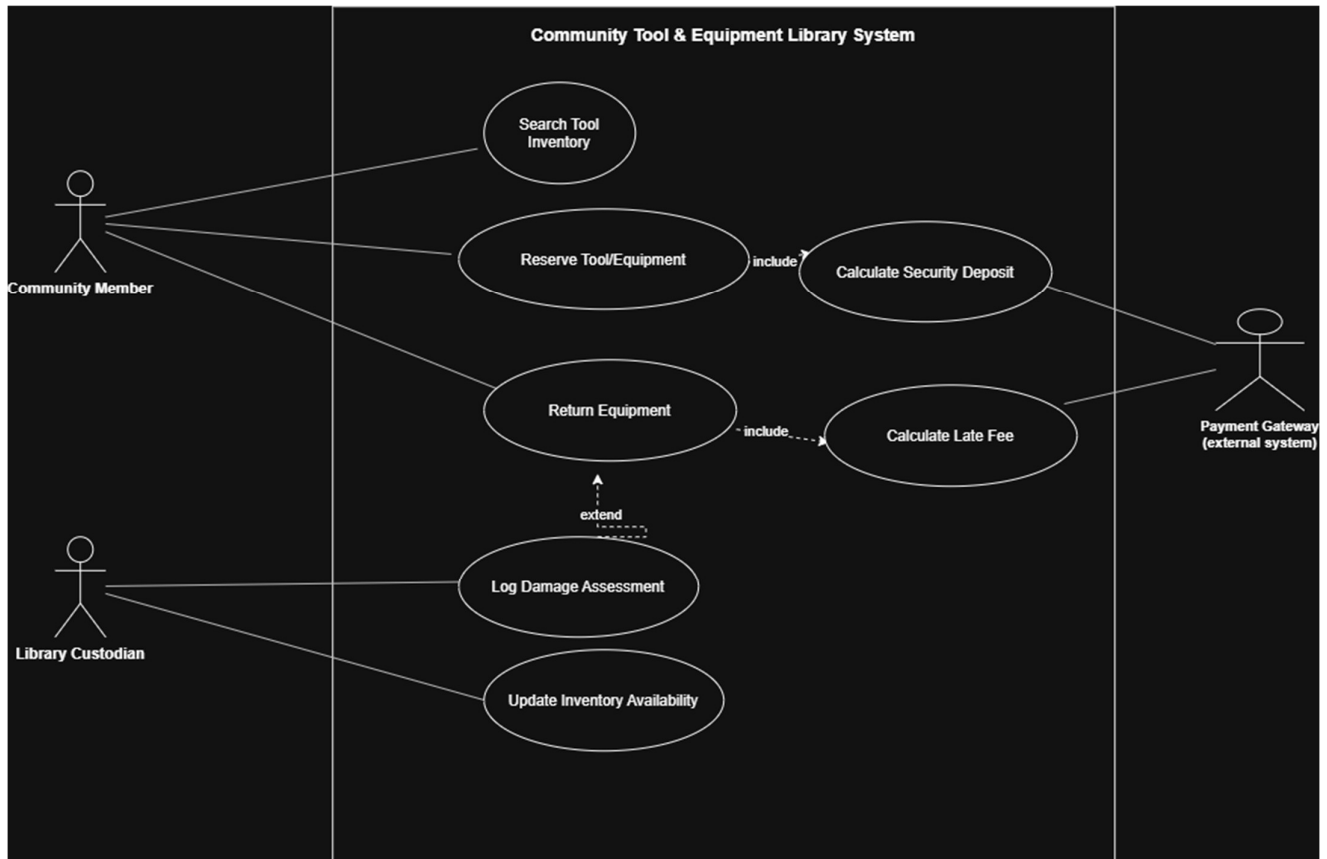
COMMUNITY TOOL AND EQUIPMENT LIBRARY

REQUIREMENTS TABLE

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall calculate required security deposits based on tool category, enforce loan duration limits, and calculate late return fees.	High	Pass: Tool reservation confirmed with deposit hold. Fail: A member with overdue tools borrows additional high-value equipment.	Protects shared assets and enforces fair borrowing policy across the community.
FR-002	Functional	The system shall allow a Community Member to search the tool/equipment inventory and view real-time availability at each neighbourhood depot locker.	High	Pass: Search results show only items currently marked "Available" and refreshed within 5 seconds of any status change. Fail: A reserved or checked-out item still appears as available.	Prevents wasted trips to a depot and double-booking of shared equipment.
FR-003	Functional	The system shall allow a Community Member to reserve an available tool for a chosen pickup depot and time window. It automatically invokes the Calculate Security Deposit function to place a deposit hold.	High	Pass: A reservation record is created with status "Confirmed" after the Payment Gateway authorizes the deposit hold. Fail: A reservation is confirmed without an authorized deposit hold.	Guarantees the library can recover costs if equipment which are not returned or damaged.
FR-004	Functional	The system shall allow the Library Custodian to log a damage assessment at return time, recording damage, photos, and an adjusted deposit deduction.	High	Pass: A damage entry marked "Moderate" or "Severe" automatically triggers a partial deposit deduction and sets the item status to "Under Repair." Fail: Damaged equipment re-enters the available pool.	Keeps the equipment catalog accurate and ensures repair costs are recovered fairly.
FR-005	Functional	The system shall send an automated reminder to a Community Member 24 hours before their loan due date and shall calculate a late fee for each full day an item remains overdue.	Medium	Pass: A member with an item due tomorrow receives exactly one reminder. If an item is returned 3 days late it shows a late fee equal to 2x the published per-day rate. Fail: No reminder is sent, or	Reduces overdue returns and keeps equipment circulating for other members.

				the late fee is miscalculated.	
NFR-001	Non-Functional	The tool inventory search must display real-time physical availability across neighborhood depot lockers.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load Fail: Latency exceeds a certain percentage or any data is transmitted unencrypted.	Members must trust that displayed availability is current, and personal/deposit data must stay protected.
NFR-002	Nonfunctional	The system shall remain available for reservations and returns during operating hours and shall present all deposit and fee amounts in a clear, unambiguous currency format.	Medium	Pass: Monthly uptime logs show at least 99% availability during operating hours, and a usability test confirms 9 of 10 members correctly interpret a deposit/fee amount unassisted. Fail: Uptime falls below 99%, or more than 1 tester misreads a fee amount.	Members depend on the system during weekday and weekend pickup windows, unclear fee displays lead to disputes.

UML DIAGRAM



ACTORS

Actor	Role
Community Member	Primary actor. Searches inventory, reserves and returns tools/equipment, pays deposits and late fees.
Library Custodian	Primary actor. Inspects returned items, logs damage assessments, and keeps depot inventory records current.
Payment Gateway	Secondary/external system actor. Authorizes deposit holds, captures deductions, and processes late-fee payments.

USECASE

USECASE	NAME	ACTOR/RELATIONSHIP
UC-01	Search Tool Inventory	Community Member
UC-02	Reserve Tool/Equipment	Community Member
UC-03	Calculate Security Deposit	«include» of UC-02 (also used by UC-04 via UC-06's payment path)
UC-04	Return Equipment	Community Member
UC-05	Log Damage Assessment	Library Custodian — «extend» of UC-04 (only when damage is found)
UC-06	Calculate Late Fee	«include» of UC-04
UC-07	Update Inventory Availability	Library Custodian

Include

UC-02 Reserve Tool/Equipment always includes UC-03 Calculate Security Deposit, since every reservation requires a deposit hold.

UC-04 Return Equipment always includes UC-06 Calculate Late Fee, since every return is checked for lateness (the fee may simply be zero).

Extend

UC-05 Log Damage Assessment optionally extends UC-04 Return Equipment: it only fires when the Library Custodian finds damage during inspection.

MAIN FLOW

1. Community Member searches the inventory and selects an available tool at a depot locker.
2. System displays tool details: category, deposit amount, maximum loan duration, and pickup depot.
3. Member selects a pickup date/time and confirms intent to reserve.
4. System invokes «include» Calculate Security Deposit, computing the deposit based on tool category.
5. System verifies the member has no outstanding overdue items .

6. System sends a deposit-hold request to the Payment Gateway.
7. Payment Gateway authorizes the hold and returns a confirmation code.
8. System creates the reservation record, sets the tool status to "Reserved," and generates a pickup code.
9. System displays a confirmation screen and emails the member the pickup code and deposit receipt.
10. Use case ends successfully.

ALTERNATE FLOW

- 6a1. The Payment Gateway declines the deposit-hold request.
- 6a2. System displays a "deposit hold failed" message and prompts the member to try another payment method.
- 6a3. If the hold fails after 2 attempts, the reservation is cancelled and the tool remains "Available."
- 6a4. Use case ends without a reservation.